Design and Analysis of Approximate Multiplier for Low-Power AI Applications

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ABSTRACT

- AI and deep learning applications are growing rapidly and need hardware that is fast, small, and uses less power. Multiplication is the most commonly used and resource-heavy operation in AI systems.
- Approximate computing allows hardware to make small, bounded errors — since AI applications remain accurate despite tiny errors, approximate multipliers help reduce power, area, and hardware complexity.
- This project designs and analyzes a 4-bit × 4-bit approximate multiplier for low-power AI applications.
- An exact multiplier is first designed in Verilog HDL using the standard partial-product method, always producing the correct answer and serving as the reference for comparison.
- An approximate multiplier is then designed by removing some smaller partial products and keeping only the important larger ones, making the circuit simpler and more power-efficient.
- Both designs are tested using Icarus Verilog and GTKWave, synthesized on Xilinx Vivado to measure LUT usage, speed, and power, and compared using Python to calculate the error.
- The results show that the approximate multiplier offers a practical trade-off between hardware savings and accuracy, making it well suited for low-power AI hardware.

Keywords: *Approximate Computing, Approximate Multiplier, AI Applications, Partial Product Truncation, Verilog HDL, Low-Power VLSI.*

EXISTING WORK

- Base paper designs a low-power approximate 8×8 unsigned multiplier for deep neural network (DNN) applications, built around a custom 4:2 compressor using NOR/NAND-based logic for a short critical path.
- The compressor makes an error in only 1 out of 256 input combinations. Synthesized in Verilog HDL on Cadence Genus (UMC 90nm), it achieves up to 30.24% energy and 33.14% power savings over existing designs, with MRED of just 0.109%.
- Validated on real DNN tasks: MNIST digit recognition using LeNet-5 (98.24% exact vs 96.45% approximate) and FFDNet image denoising — confirming the approximation barely affects DNN performance.
- **Research Gap**: this accuracy needs a fully custom-redesigned compressor (Area 30.57 μm^2, Delay 237ps) verified on a commercial EDA flow (Cadence Genus, 90nm) — high design and verification cost, not achievable with open-source tools. An 8×8 design also needs exhaustive verification across 65,536 input combinations, which is heavy for open-source simulation flows — so this project targets a 4×4 multiplier, enabling full exhaustive verification (all 256 combinations) while validating the same truncation-based approximation technique, extensible to 8×8 and higher-order multipliers as future work.
- My Project achieves a similar trade-off with simple partial-product truncation: 37.5% LUT reduction, 17.0% delay reduction, and 45.7% power reduction over an exact multiplier — verified entirely on free, open-source tools (Icarus Verilog, GTKWave, Vivado WebPACK).

SOFTWARE TOOLS

- VS Code (Latest stable) — RTL and Python code editing.
- Icarus Verilog (10.3) — Verilog compilation and simulation.
- GTKWave (3.3.103) — Waveform inspection (VCD).
- Python 3 (3.8.10) with NumPy / Matplotlib (Latest) — Error analysis, numerical analysis, and plotting.
- Vivado WebPACK (2026.1) — Synthesis and timing/power reports. Installed via apt (Icarus Verilog, GTKWave, Python) and pip (NumPy, Matplotlib, pandas); verified with iverilog -V, gtkwave --version, and python3 --version.

EXACT MULTIPLIER — PARTIAL PRODUCT METHOD

- A 4-bit × 4-bit multiplication generates four partial products (pp0–pp3), each formed by shifting A according to a bit of B.
- All four partial products are summed to produce the 8-bit result — mathematically perfect for every input.
- Verified across all 256 possible 4-bit input combinations — zero error, 100% exhaustive coverage.
- Worked Example: 10 × 12 = 120 → pp2 = 00101000, pp3 = 01010000, Sum = 01111000 = 120.
- Serves as the golden reference used to measure the accuracy and hardware savings of the approximate design.

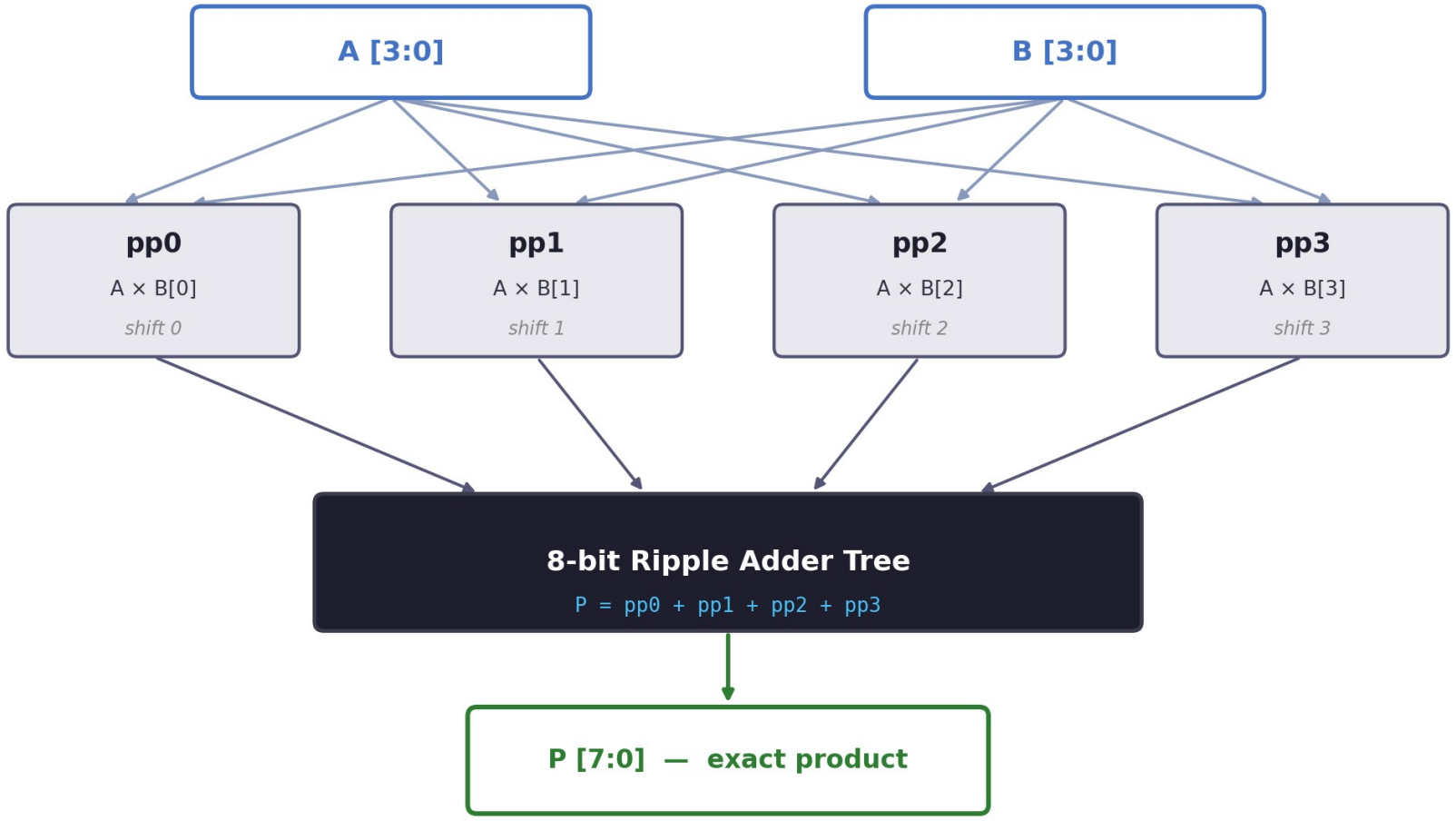
exact_multiplier.v

```
module exact_multiplier (
  input  [3:0] A, B,
  output [7:0] P
);

wire [7:0] pp0 = B[0] ?
  {4'b0, A} : 8'b0;
wire [7:0] pp1 = B[1] ?
  {3'b0, A, 1'b0} : 8'b0;
wire [7:0] pp2 = B[2] ?
  {2'b0, A, 2'b0} : 8'b0;
wire [7:0] pp3 = B[3] ?
  {1'b0, A, 3'b0} : 8'b0;

assign P = pp0+pp1+pp2+pp3;
endmodule
```

EXACT MULTIPLIER — BLOCK DIAGRAM



All four partial products fully summed — zero error across all 256 input combinations

APPROXIMATE MULTIPLIER — PARTIAL-PRODUCT TRUNCATION

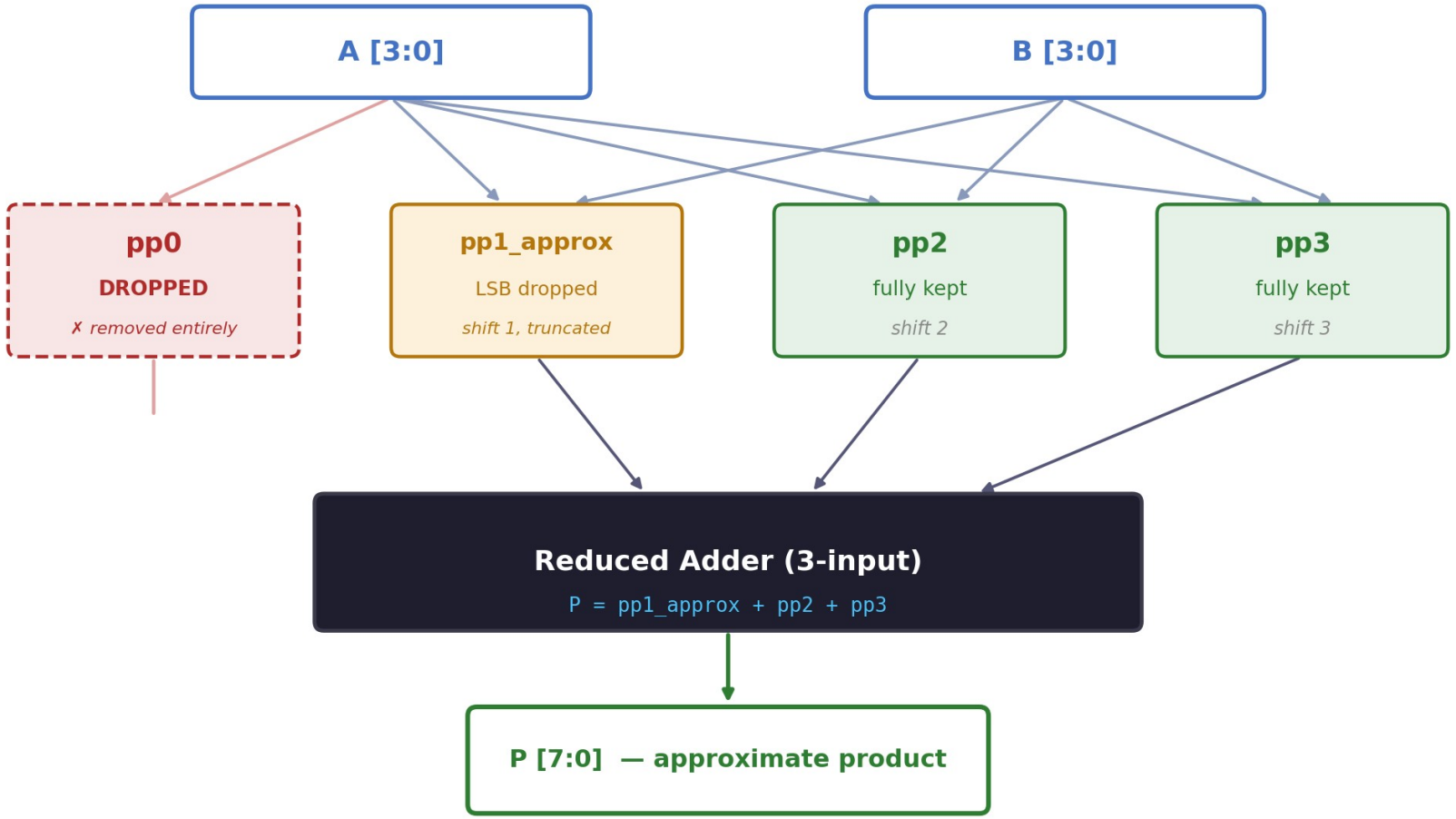
- The approximate multiplier drops pp0 entirely and truncates pp1, keeping pp2 and pp3 fully intact.
- pp3 & pp2: fully kept | pp1: truncated (LSB dropped) | pp0: dropped entirely.
- Removing one full-adder level shortens the critical path, reduces LUT usage, and lowers switching power.
- Both the exact and approximate multipliers share a common testbench so their outputs can be directly compared.

approx_multiplier.v

```
wire [7:0] pp3 = B[3] ?
  {1'b0,A,3'b0} : 8'b0;
wire [7:0] pp2 = B[2] ?
  {2'b0,A,2'b0} : 8'b0;
wire [7:0] pp1_approx = B[1] ?
  {3'b0,A[3:1],2'b0} : 8'b0;

// pp0 dropped
assign P = pp1_approx
         + pp2 + pp3;
```

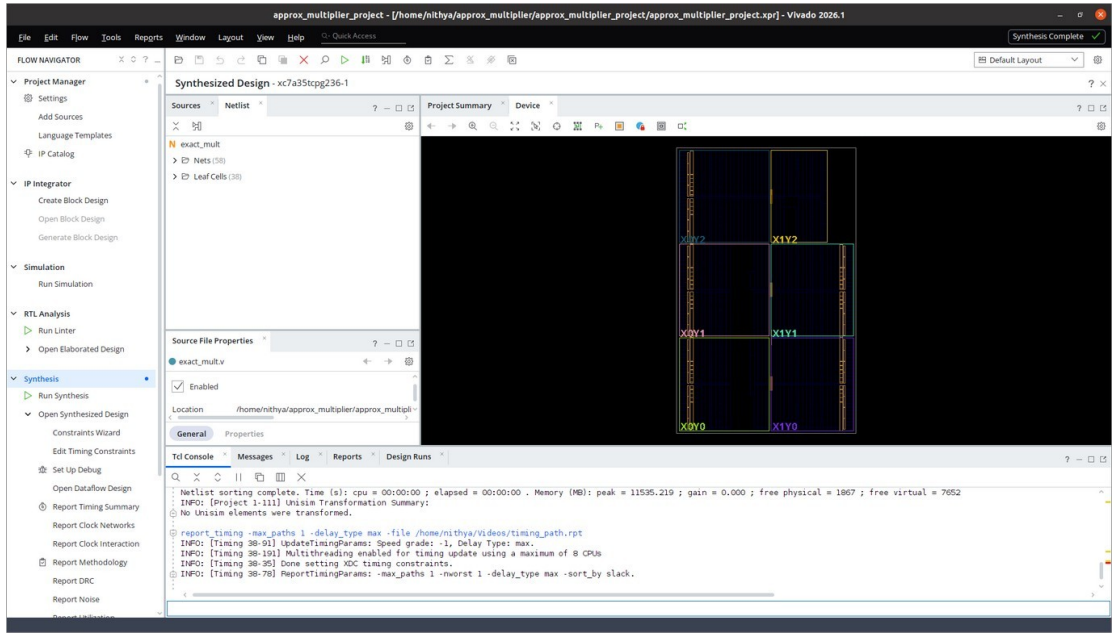
APPROXIMATE MULTIPLIER — BLOCK DIAGRAM



One full-adder level removed → shorter critical path, fewer LUTs, lower switching power

SYNTHESIS OUTPUTS

Vivado synthesized-design view — Exact Multiplier (Xilinx Artix-7, xc7a35tcpg236-1)

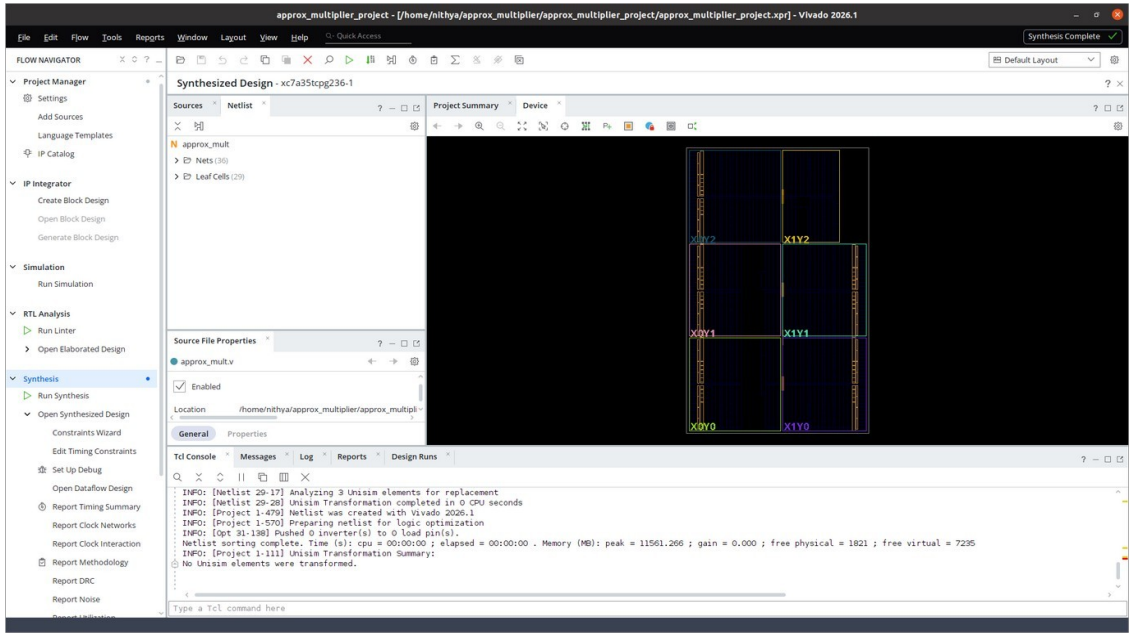


Exact Multiplier — Synthesized

- Leaf cells: 38 — synthesized cell count for the exact multiplier design.
- Nets: 58 — interconnect nets in the exact multiplier design.

SYNTHESIS OUTPUTS

Vivado synthesized-design view — Approximate Multiplier (Xilinx Artix-7, xc7a35tcpg236-1)

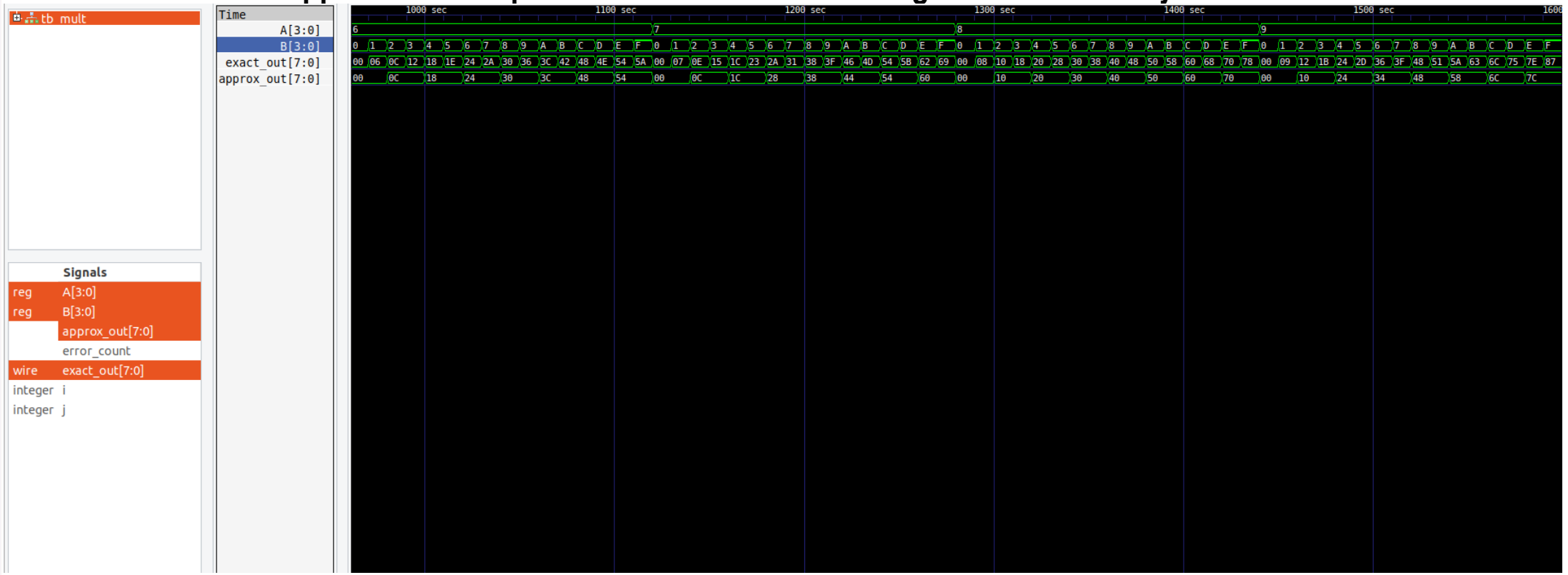


Approximate Multiplier — Synthesized

- Leaf cells: 29 — fewer synthesized cells confirm the reduced logic footprint (vs 38 for exact).
- Nets: 36 — simplified partial-product interconnect (vs 58 for exact).

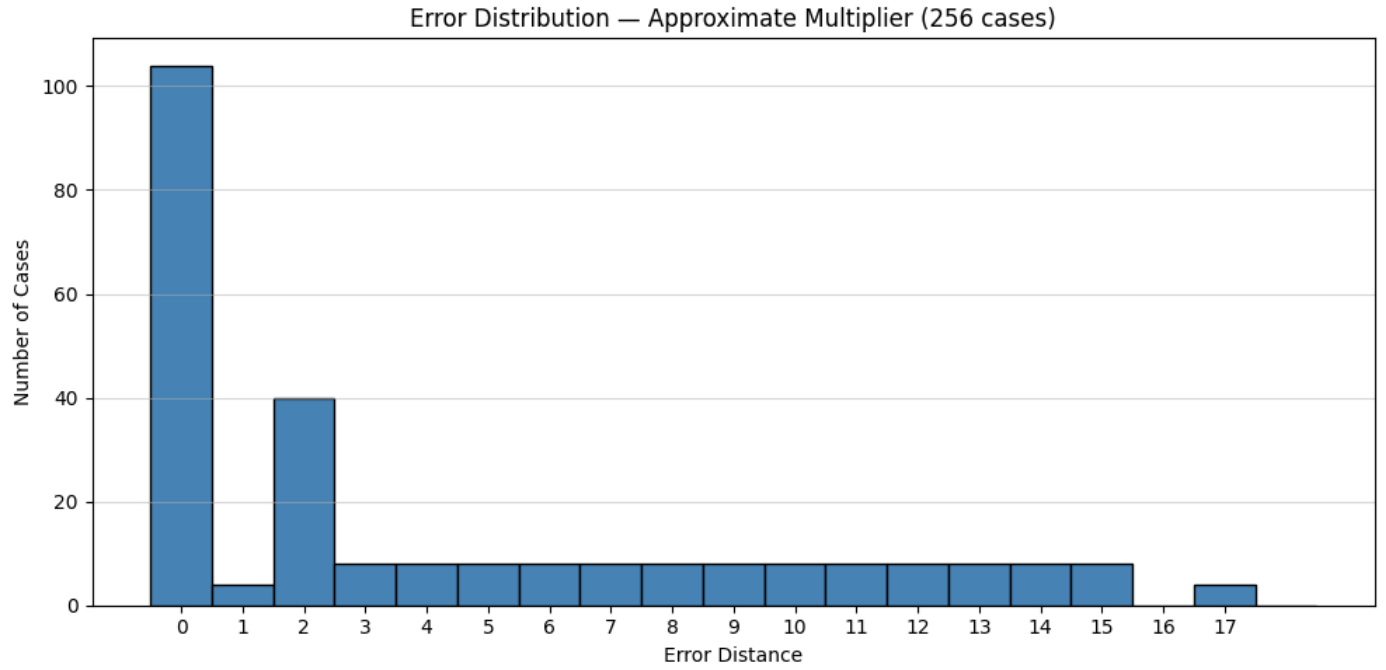
SIMULATION RESULTS — GTKWAVE VERIFICATION

Common testbench applies all 256 input combinations to both designs simultaneously



- Observation: Outputs match exactly for (3,5), (4,6), (6,6) — dropped bits don't contribute. Visible divergence appears for (1,8), (7,2), (15,15), (9,3), (12,11).

ERROR ANALYSIS — PYTHON METRICS



4.25
Mean Error Distance (MED)

59.38%
Error Rate (ER)

17
Max Error Distance

16.12%
Mean Relative Error (MRE)

- Errors increase with operand magnitude but remain bounded (Max Error Distance = 17) — a predictable pattern that AI workloads can tolerate.
- Max error (17) occurs at the largest operand pairs: (A=15, B=3), (15, 7), (15, 11), (15, 15).

SYNTHESIS RESULTS — HARDWARE COMPARISON

Synthesis results from Vivado 2026.1, target device xc7a35tcpg236-1 (Artix-7).

Metric	Exact	Approximate	Savings
LUTs	16	10	**37.5%**
Critical Path Delay (ns)	6.990	5.804	**17.0%**
Total On-Chip Power (W)	4.717	2.563	**45.7%**

- LUT count and Total On-Chip Power taken from report_utilization and report_power on the synthesized design.
- Critical Path Delay is the Data Path Delay from report_timing (max path), since both designs are purely combinational with no clock defined.
- The approximate multiplier cuts LUT usage by 37.5%, delay by 17.0%, and power by 45.7% compared with the exact design.

APPLICATIONS

- **Edge-AI Inference Accelerators —** smart cameras and surveillance systems performing on-device object detection (e.g., person or vehicle recognition) need fast, low-power hardware to avoid constant cloud offloading, extending battery life for always-on operation.
- **Low-Power IoT & Wearable Devices —** fitness bands and smartwatches running on-device health analytics, such as heart-rate anomaly or fall detection, operate on tiny batteries for days at a time, so every milliwatt saved in the multiplier directly extends usable battery life.
- **Real-Time Signal Processing —** voice-assistant wake-word detection (e.g., “Hey Siri”, “OK Google”) on hearables and smart speakers must continuously process audio streams within strict power and latency budgets, making approximate multipliers well suited for always-listening front ends.
- **Mobile Neural-Network Hardware —** smartphone NPUs handling everyday on-device AI tasks, such as face unlock, camera scene detection, and live translation, rely on millions of multiply operations per second; approximate multipliers cut power without noticeably affecting user-facing accuracy.

CONCLUSION & FUTURE WORK

Conclusion:

- Exact multiplier: zero error across all 256 test cases (golden reference).
- Approximate design: 37.5% fewer LUTs, 17.0% less delay, 45.7% less power.
- Error stays bounded (Max Error Distance = 17) and data-dependent.
- AI matrix-multiply demo confirms inference quality is still acceptable.
- **A small, controlled error buys big gains in speed, size, and power.**

Future Work:

- Scale the design to 8×8 and 16×16 multipliers.
- Explore signed multiplication and other approximate techniques.
- Validate the design on real FPGA hardware.

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THANK YOU